



Mil-Spec / ITAR Considerations

Military, defense, and export-controlled hardware

The rules that apply when your PCB has a weapons application

Audience: Engineers working in defense / aerospace

MIL-STD • ITAR • EAR • Conformal coat • Domestic sourcing



What is 'Mil-Spec' Hardware

Performance

Operates -55 to +125 °C
Guaranteed at extremes.

Reliability

MTBF measured in years,
not months.

Robustness

Vibration, shock, EMI
survival.

Documentation

Every component traceable
to manufacturer + batch.

Sourcing

US-domestic or allied-country
for sensitive items.

Certification

MIL-STD-883 + AS 9100 + JEDEC
combo standards.

Key Military Standards

- ▶ MIL-STD-883: test methods for microelectronics
- ▶ MIL-STD-810: environmental test conditions (vibration, shock, humidity)
- ▶ MIL-STD-461: EMI / EMC requirements
- ▶ MIL-PRF-31032: rigid printed circuit boards qualification
- ▶ MIL-PRF-50884: flex / rigid-flex qualification
- ▶ MIL-STD-202: passives test methods
- ▶ Combined with: AS 9100 / IATF 16949 / IPC standards
- ▶ Often customer-specific specs override these (e.g., Lockheed-specific reqs)

ITAR — International Traffic in Arms

- ▶ ITAR = USA export control for defense articles + services
- ▶ Administered by State Department (Directorate of Defense Trade Controls — DDTC)
- ▶ Items on USML (US Munitions List)
- ▶ Examples: military communications equipment, weapons systems, related software/hardware
- ▶ Penalties: criminal — fines up to \$1M per violation + imprisonment
- ▶ Compliance: ITAR registration (\$2,250+/year), classification of every product
- ▶ Hardware containing ITAR-controlled IP cannot ship outside USA without license
- ▶ If you're outside USA: cannot work on ITAR-controlled products without authorisation

EAR — Export Administration Regulations

- ▶ EAR = USA dual-use export controls (civilian items with military potential)
- ▶ Administered by Department of Commerce (BIS — Bureau of Industry and Security)
- ▶ Items on CCL (Commerce Control List), categorised by ECCN codes
- ▶ Includes: encryption, sensors, advanced manufacturing, semiconductors > 22nm
- ▶ Most consumer electronics fall under EAR99 (relatively unrestricted)
- ▶ Some items require export license depending on destination country
- ▶ Embargo lists: certain countries (North Korea, Iran, Syria, etc.) forbidden
- ▶ Many SoCs (NVIDIA, AMD high-end) fall under EAR — restricted to certain countries

Indian Context

- ▶ India ≠ USA — ITAR/EAR are USA-specific
- ▶ But: many India-based defense projects use USA-sourced components → must comply with USA export controls
- ▶ India's equivalent: Foreign Trade (Development and Regulation) Act + SCOMET list
- ▶ SCOMET: Special Chemicals, Organisms, Materials, Equipment + Technologies
- ▶ DGFT (Directorate General of Foreign Trade) administers Indian export control
- ▶ DRDO, BEL, HAL: India's domestic defense electronics companies
- ▶ Working with foreign primes (e.g., Boeing India, Airbus India): must follow USA + EU controls
- ▶ Indian classification: 'Make in India' defense electronics is growing fast

Conformal Coating

- ▶ Conformal coat: thin polymer layer over assembled PCB
- ▶ Common types: acrylic, urethane, silicone, parylene, epoxy
- ▶ Protects from: humidity, dust, salt fog, chemicals, fungi
- ▶ Mil-spec: typically Type AR (acrylic) or Type UR (urethane) per IPC-CC-830
- ▶ Thickness: 25-75 μm typical
- ▶ Applied via: brush, spray, dip, or vapor deposition (parylene)
- ▶ Cost: ₹50-200 per board depending on coat + complexity
- ▶ Test points + connectors masked off before coating

Domestic Sourcing

- ▶ DPAS (Defense Priority Allocations System) — USA defense priority programs
- ▶ Domestic content requirements (Buy American) for many DoD contracts
- ▶ Trusted Foundry programs — DoD-qualified semiconductor manufacturers
- ▶ Counterfeit electronics: major concern; mil-spec requires audited supply chain
- ▶ Component traceability: lot-level tracking required
- ▶ India: 'Make in India' for defense — preference for domestic production
- ▶ Domestic Indian PCB fabs growing: AT&S India, Centum, BEL
- ▶ Foreign components require government clearance for many India defense projects

Environmental Testing

- ▶ MIL-STD-810 covers wide range of environmental conditions
- ▶ Temperature: -55 to +125 °C operating, broader storage
- ▶ Humidity: 95% RH cycling
- ▶ Vibration: 5-2000 Hz random spectrum
- ▶ Mechanical shock: 40 g shock, 11 ms duration
- ▶ Salt fog: 1000 hours exposure
- ▶ Altitude: low-pressure operation
- ▶ Fungus growth: tropical conditions test

Reliability + Lifecycle

- ▶ Mil-spec products designed for 20-30 year lifecycles
- ▶ Parts EOL planning critical — secure stocks of obsolete parts
- ▶ Reverse-engineering kept as documentation for late-life rework
- ▶ Schematic + layout files preserved indefinitely
- ▶ Spare-parts planning: 10-20% of fielded units worth of spares
- ▶ Field maintenance manuals + tools
- ▶ BIT (Built-In Test) firmware mandatory
- ▶ Component derating: 50-60% of rated power (vs 70% commercial)

Common Issues

- ▶ Using commercial-grade components in mil-spec product (won't survive temp/vibration)
- ▶ Working with foreign engineers on ITAR-controlled designs without authorization
- ▶ Buying components from grey market — counterfeit risk
- ▶ Missing conformal coat — humidity damage in field
- ▶ Not respecting BIT requirements — system can't self-diagnose
- ▶ Insufficient component derating — early field failures
- ▶ Long lifecycle assumed but parts EOL prematurely → expensive last-time-buy
- ▶ Documentation lost — can't repair fielded units after years

Working on Mil/Defense Projects in India

- ▶ Indian companies: DRDO, BEL, HAL, Tata Advanced Systems, L&T Defence
- ▶ Government approvals via DGFT + Defence Ministry for sensitive projects
- ▶ If foreign IP involved: ITAR/EAR + India's classification
- ▶ Make in India initiative: preference for domestic + indigenous content
- ▶ PLI scheme: government incentives for domestic electronics manufacturing
- ▶ Compliance: SCOMET classifications for export-controlled items
- ▶ Skills: AS 9100 + mil-spec audits + classified-material handling training
- ▶ Career: defense electronics is hiring strongly in India (2026 outlook)

Key Takeaways

Key takeaways from this lecture

Mil-spec = harder spec

-55 to +125 °C, full vib/shock.

ITAR = USA defense export

Strict; criminal penalties.

EAR = USA dual-use

Some commercial items restricted.

India SCOMET

Domestic export-control list.

Trace components

Lot-level for 20+ years.

Conformal coat + derate

Reliability cornerstones.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ MIL-STD-883 / 810 / 461 / 202 — DoD standards
- ▶ ITAR @ State Department (pmddtc.state.gov)
- ▶ EAR @ Commerce Department (bis.doc.gov)
- ▶ India DGFT — Foreign Trade Policy
- ▶ IPC-CC-830 — Conformal Coating standard
- ▶ Indian defense companies: DRDO, BEL, HAL careers pages
- ▶ AS 9100 + IPC standards (covered in deck 35 + 55)
- ▶ Trusted Foundry program (USA DoD)

Thank you • Build something this week.